PATENT A

Self-Sustaining Neural-Motor Energy Harvesting Loop■for Autonomous Cognitive Systems

8 drawing sheets

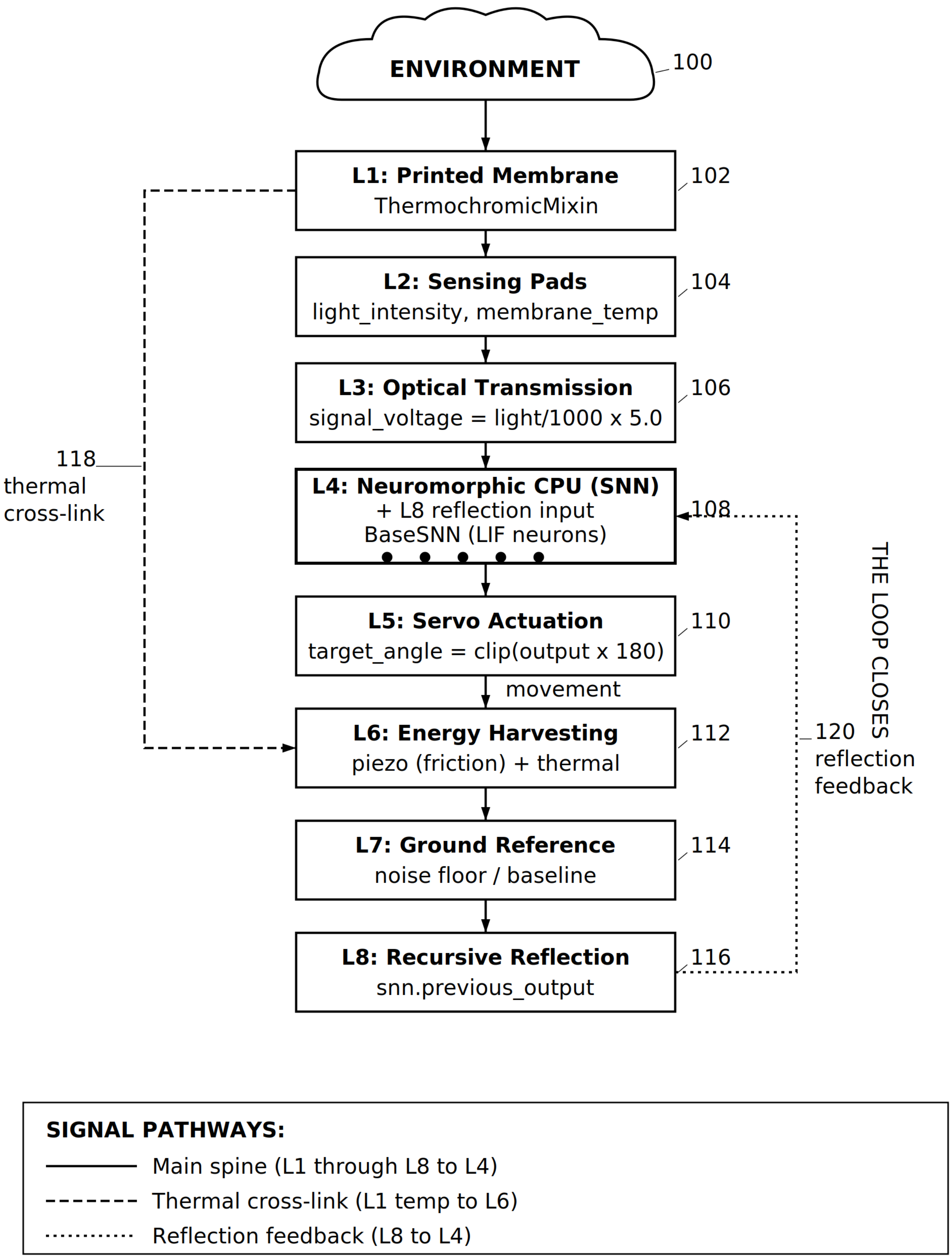

ENVIRONMENT
100
L1: Printed Membrane
ThermochromicMixin
102
L2: Sensing Pads
light_intensity, membrane_temp
104
L3: Optical Transmission
signal_voltage = light/1000 x 5.0
106
L4: Neuromorphic CPU (SNN)
+ L8 reflection input
BaseSNN (LIF neurons)
108
L5: Servo Actuation
target_angle = clip(output x 180)
110
movement
L6: Energy Harvesting
piezo (friction) + thermal
112
L7: Ground Reference
noise floor / baseline
114
L8: Recursive Reflection
snn.previous_output
116
118
thermal
cross-link
120
reflection
feedback
THE LOOP CLOSES
SIGNAL PATHWAYS:
Main spine (L1 through L8 to L4)
Thermal cross-link (L1 temp to L6)
Reflection feedback (L8 to L4)

FIG. 1

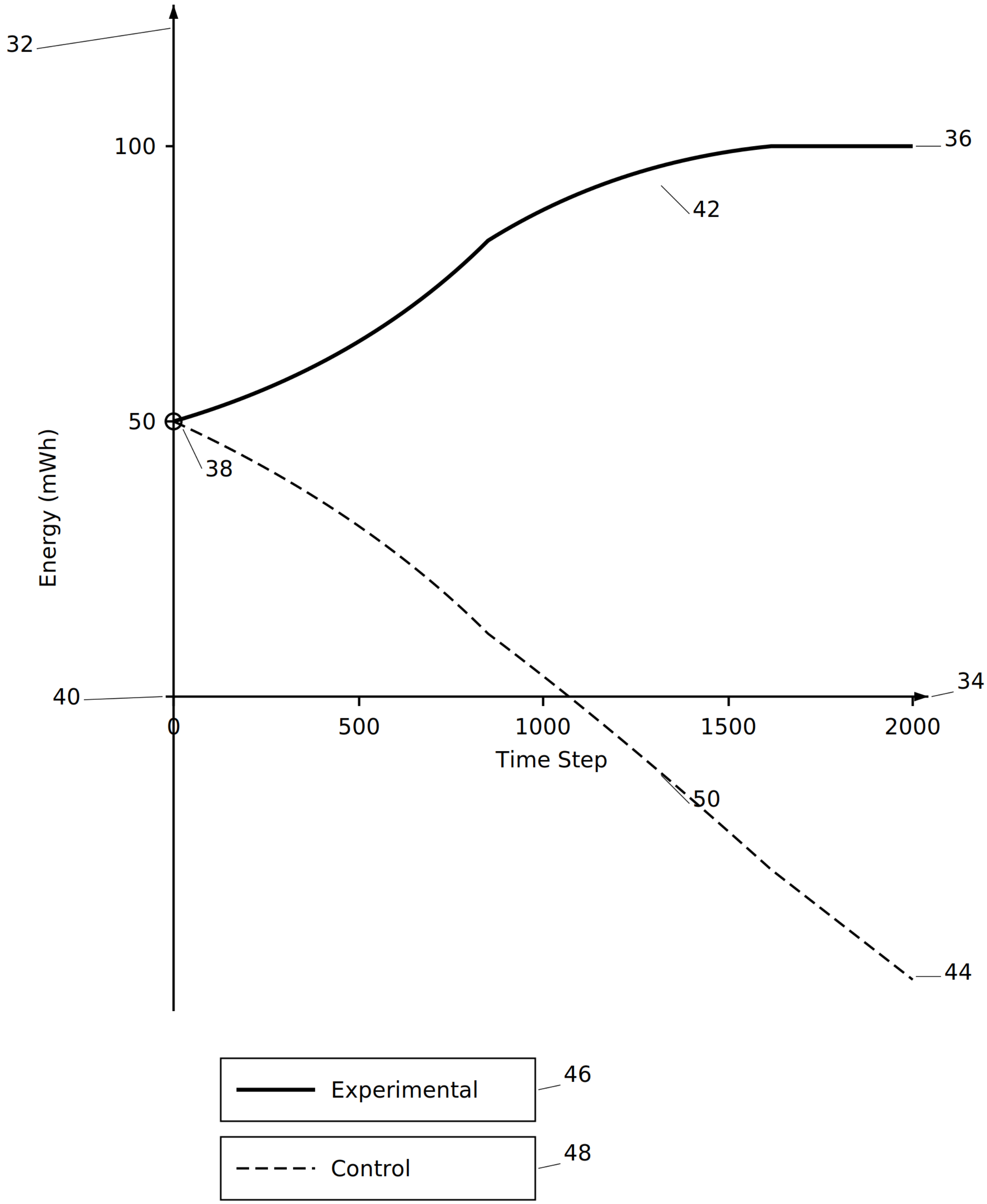

32
100
36
42
50
38
Energy (mWh)
40
34
0
500
1000
1500
2000
Time Step
50
44
Experimental
46
Control
48

FIG. 2

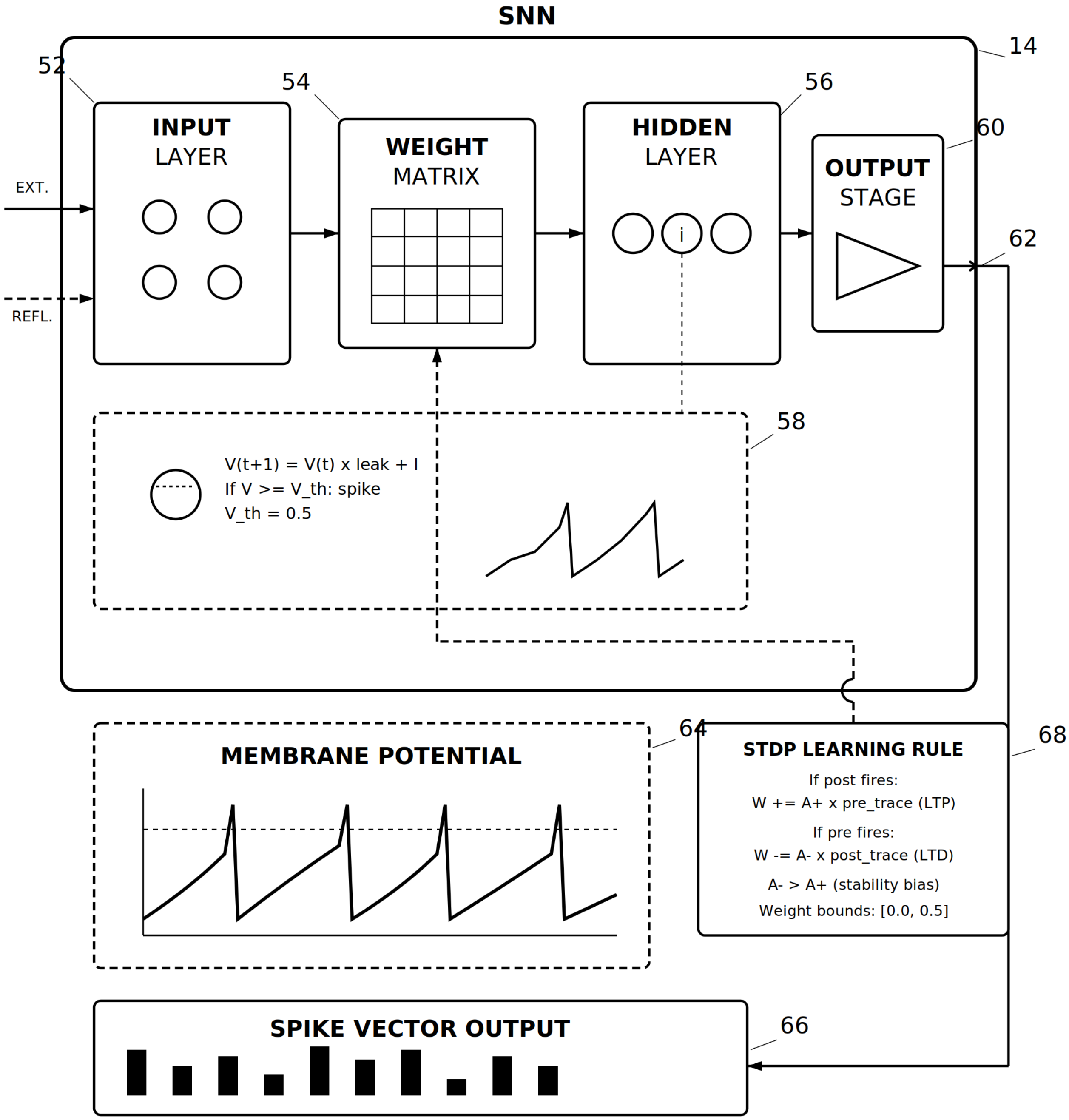

SNN
14
52
54
56
60
62
58
INPUT LAYER
WEIGHT MATRIX
HIDDEN LAYER
OUTPUT STAGE
EXT.
REFL.
i
V(t+1) = V(t) x leak + I
If V >= V_th: spike
V_th = 0.5
64
MEMBRANE POTENTIAL
STDP LEARNING RULE
If post fires:
W += A+ x pre_trace (LTP)
If pre fires:
W -= A- x post_trace (LTD)
A- > A+ (stability bias)
Weight bounds: [0.0, 0.5]
68
SPIKE VECTOR OUTPUT
66

FIG. 3

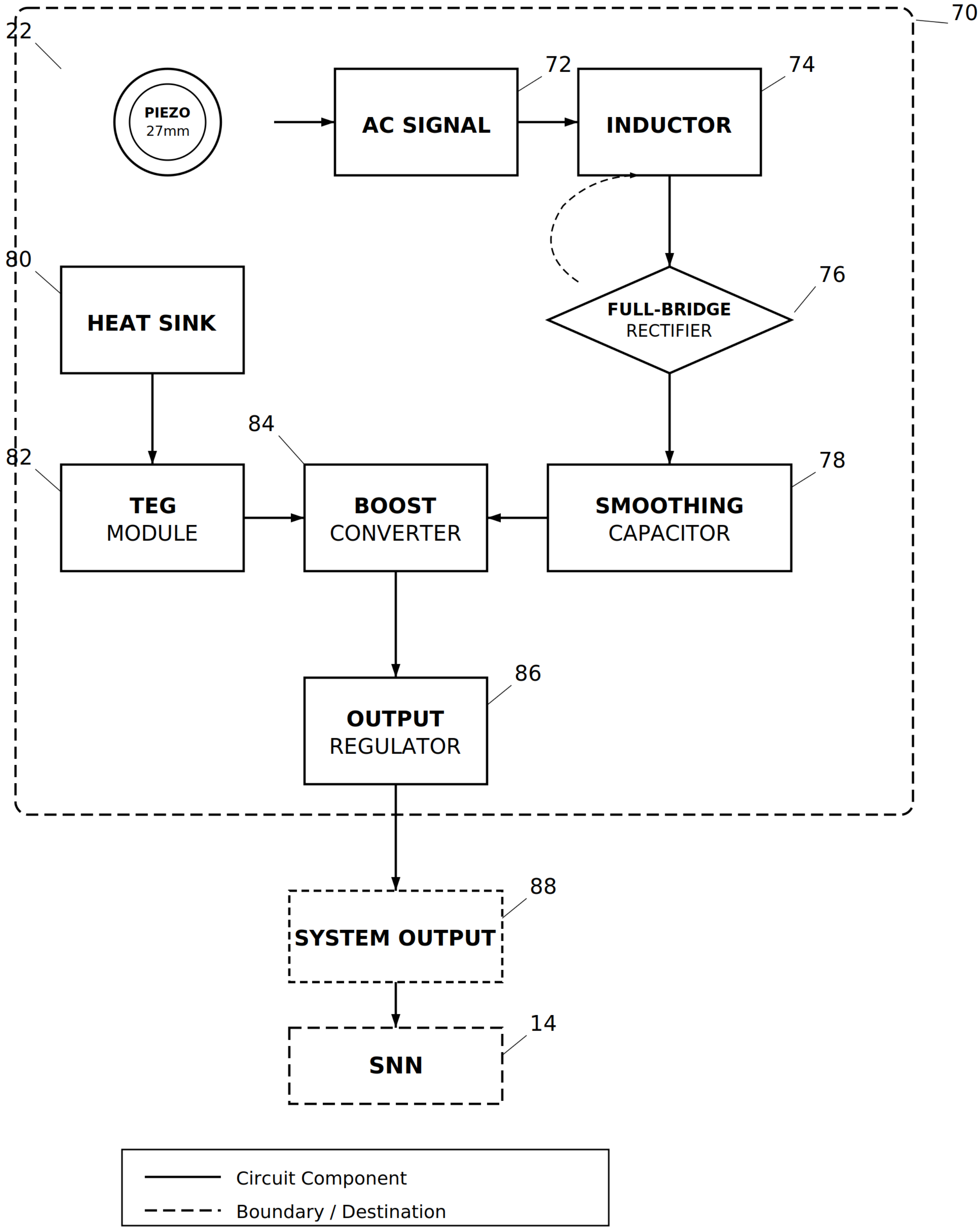
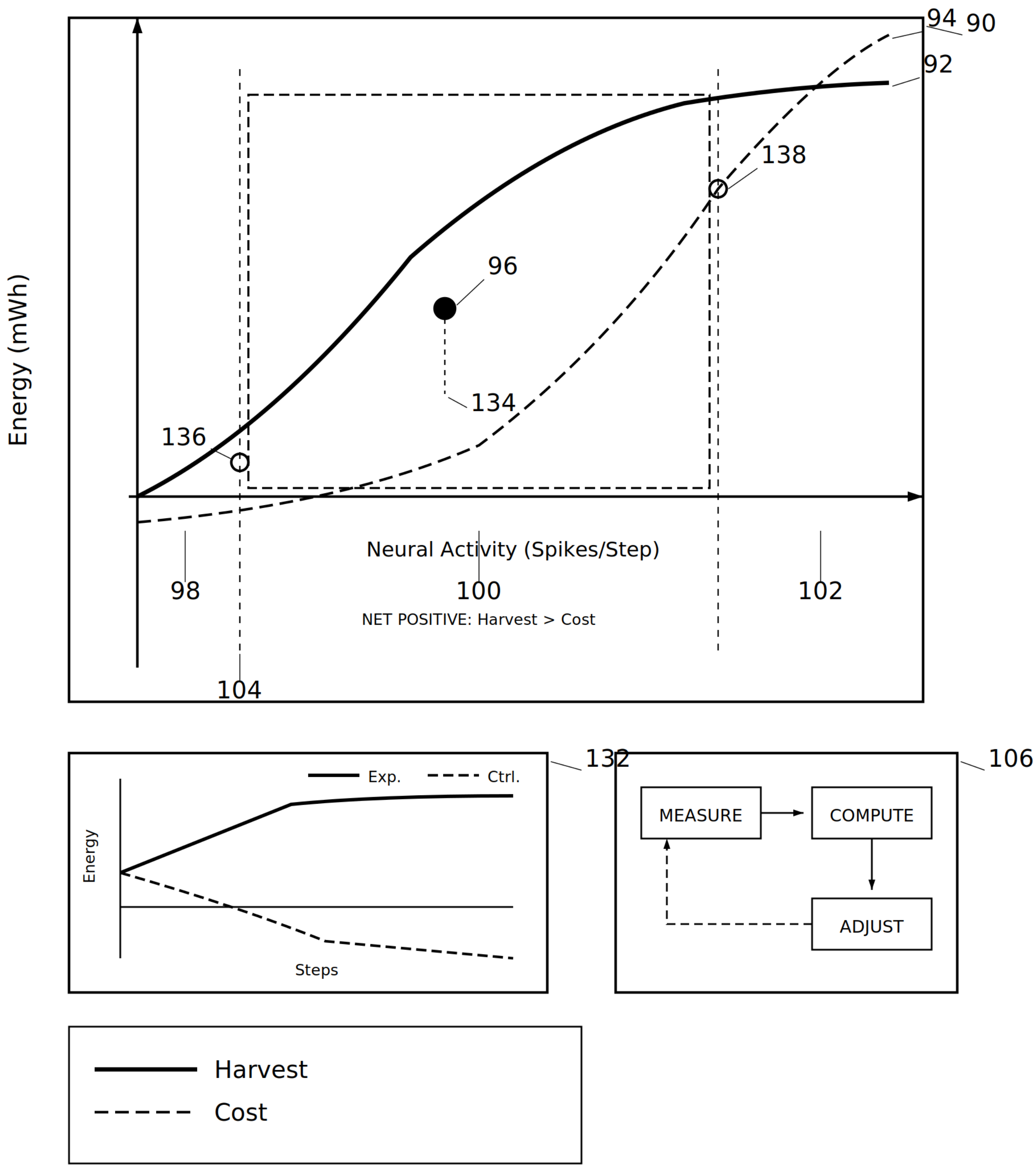

70
22
PIEZO
27mm
AC SIGNAL
72
INDUCTOR
74
80
HEAT SINK
FULL-BRIDGE
RECTIFIER
76
84
82
TEG
MODULE
BOOST
CONVERTER
SMOOTHING
CAPACITOR
78
OUTPUT
REGULATOR
86
SYSTEM OUTPUT
88
SNN
14
Circuit Component
Boundary / Destination

FIG. 4



Energy Formulas

Harvest = 0.20 + (0.09 x spikes) mWh/step

Cost = 0.225 + (0.006 x s) + (0.0003 x s^2) mWh/step

Net = Harvest - Cost

FIG. 5

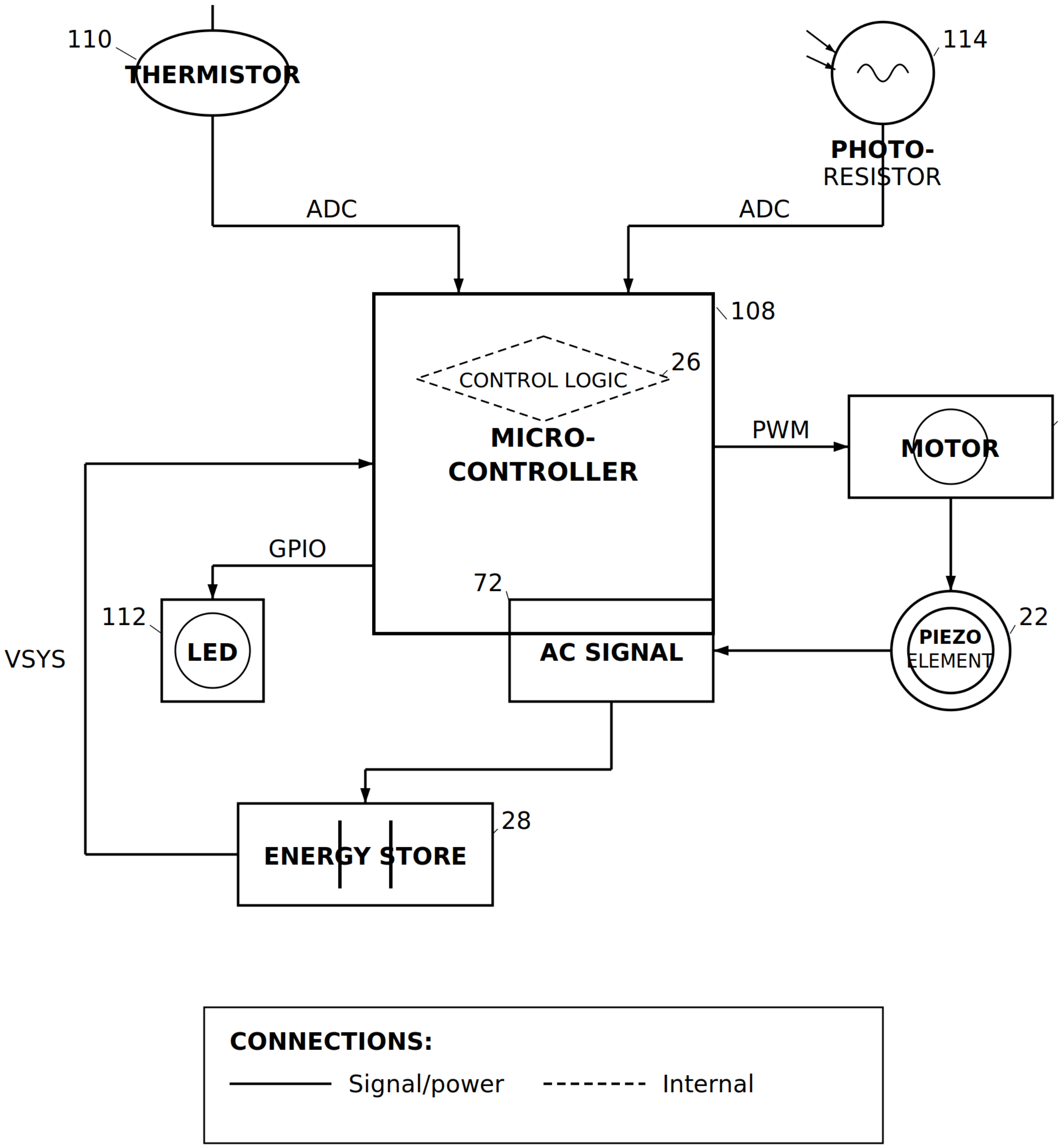

110
THERMISTOR
114
PHOTO-
RESISTOR
ADC
ADC
108
CONTROL LOGIC
26
MICRO-
CONTROLLER
PWM
MOTOR
16
GPIO
72
112
LED
VSYS
AC SIGNAL
PIEZO
ELEMENT
22
ENERGY STORE
28
CONNECTIONS:
Signal/power
Internal

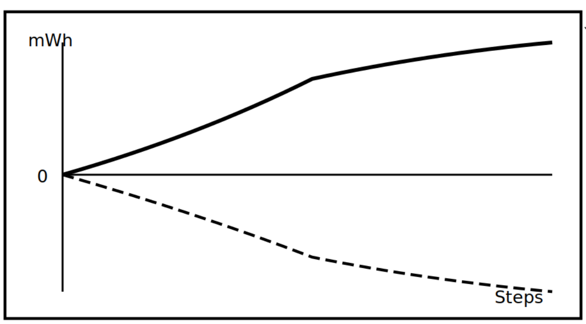
FIG. 6

116

Claim	Criterion	Measured	Threshold	Result
Phase 7 : Control Baseline				
A	Continuity	2000 pts	2000 pts	**PASS**
B	Boundedness	finite	finite	**PASS**
C	Robustness	std = 24.8	std <= 45	**PASS**
D	I/O gain	corr = 0.42	corr >= 0.2	**PASS**
E	Saturation	0.03	<= 0.20	**PASS**
Phase 8 : Synaptic Plasticity (STDP)				
F8.1	Weight entropy	0.94 bit drop	decrease	**PASS**
F8.2	STDP MI ratio	6.52x	> 1.20x	**PASS**
F8.3	Convergence	0.00%	< 10%	**PASS**
Phase 9 : Predictive Processing				
F9.1	Error reduction	63.9%	> 50%	**PASS**
F9.2	Random trend	p = 0.69	p > 0.05	**PASS**
F9.3	Adaptation	1.51x spike	> 1.3x + recover	**PASS**
Phase 10 : Multi-Modal Integration				
F10.1	Sync difference	0.70	> 0.20	**PASS**
F10.2	Weight entropy	2.31 to 0.0	decrease	**PASS**
F10.3	Convergence	0.18	< 0.20	**PASS**

118

Configuration	Energy at 2000 Steps	Status
Control (EnergyConfig)	-4,697 mWh	Depleted
Experimental (BalancedConfig)	100.0 mWh	Sustained



120

122

Experimental (sustained)

Control (depleted)

124

14 / 14 CLAIMS PASS : Phases 7 - 10 Validated

FIG. 7

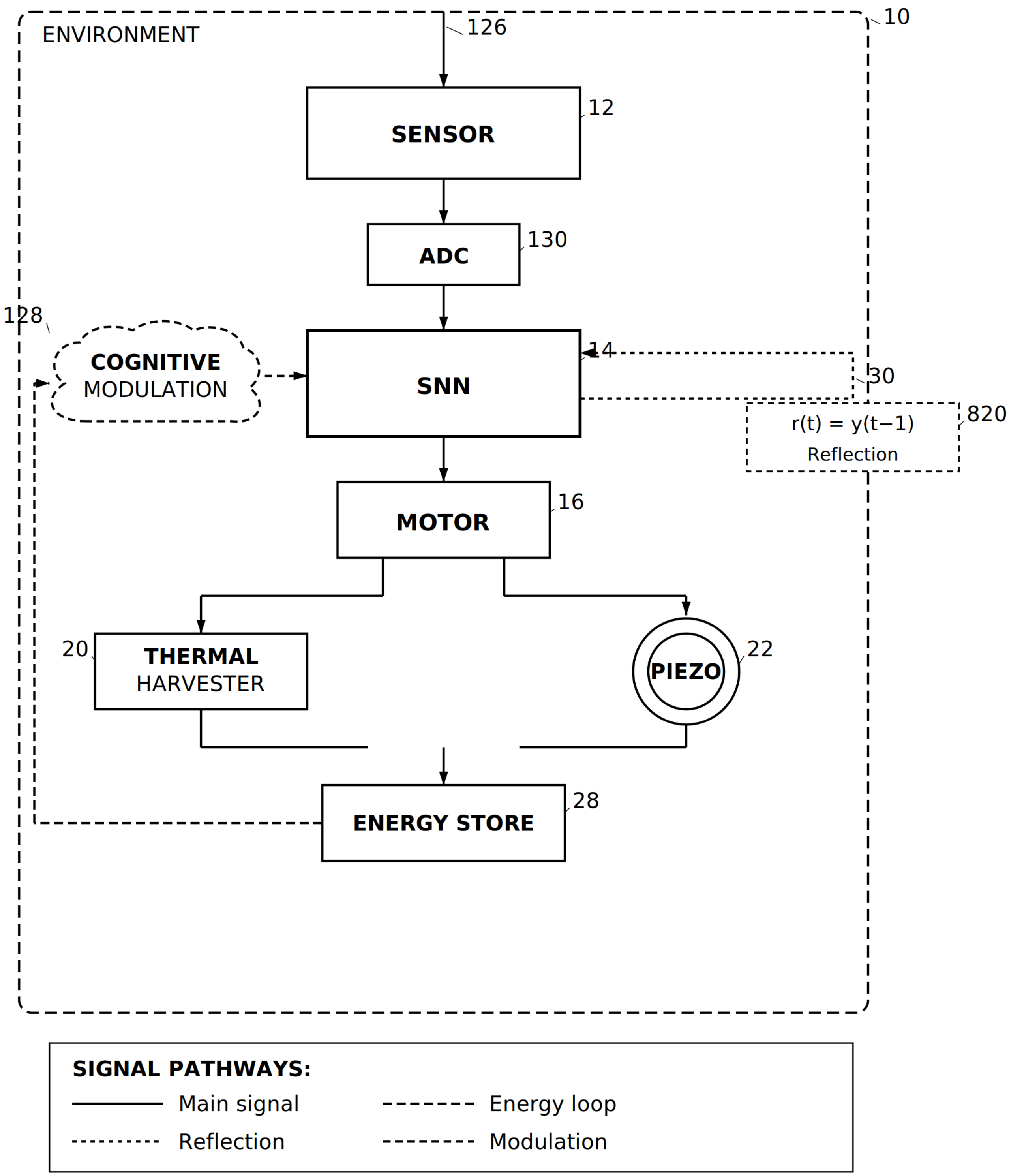

ENVIRONMENT
10
126
SENSOR
12
ADC
130
128
COGNITIVE
MODULATION
SNN
14
30
r(t) = y(t−1)
Reflection
820
MOTOR
16
THERMAL
HARVESTER
20
PIEZO
22
ENERGY STORE
28
SIGNAL PATHWAYS:
Main signal
Energy loop
Reflection
Modulation

FIG. 8

PATENT B

e Recursive Self-Observation Method■and System for Spiking Neural Networks■with Dynamic Reflection Coefficien

6 drawing sheets

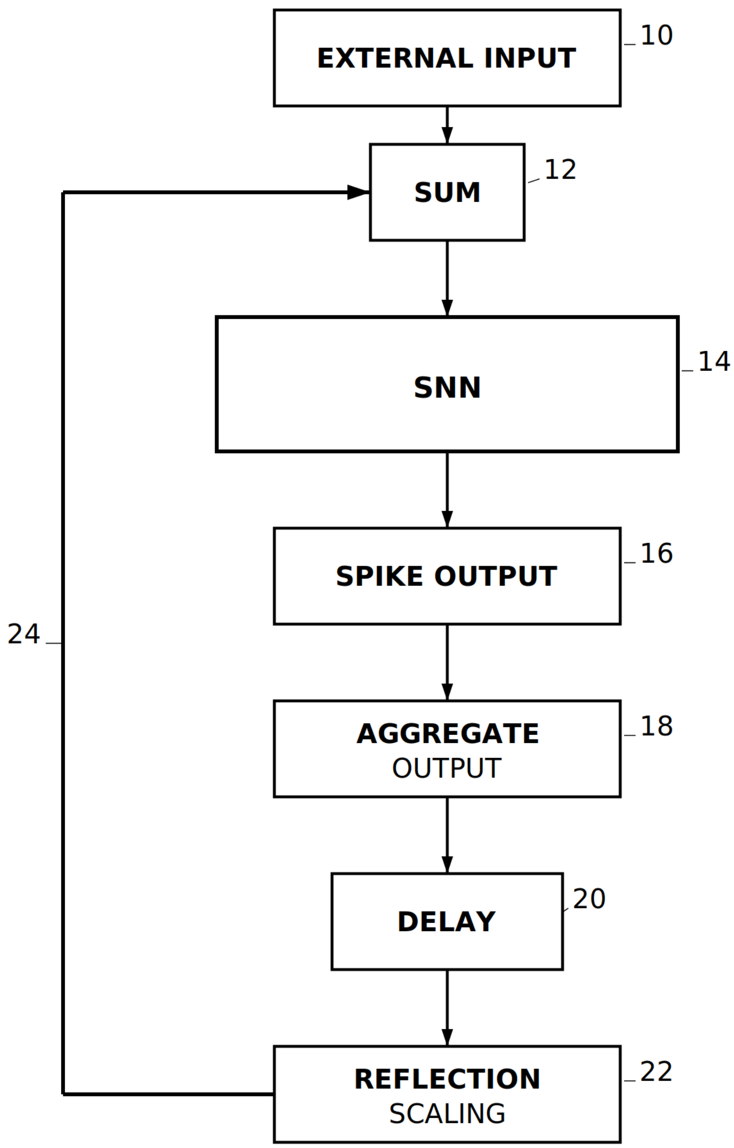

EXTERNAL INPUT
10
SUM
12
SNN
14
SPIKE OUTPUT
16
AGGREGATE
OUTPUT
18
DELAY
20
REFLECTION
SCALING
22
24

FIG. 1

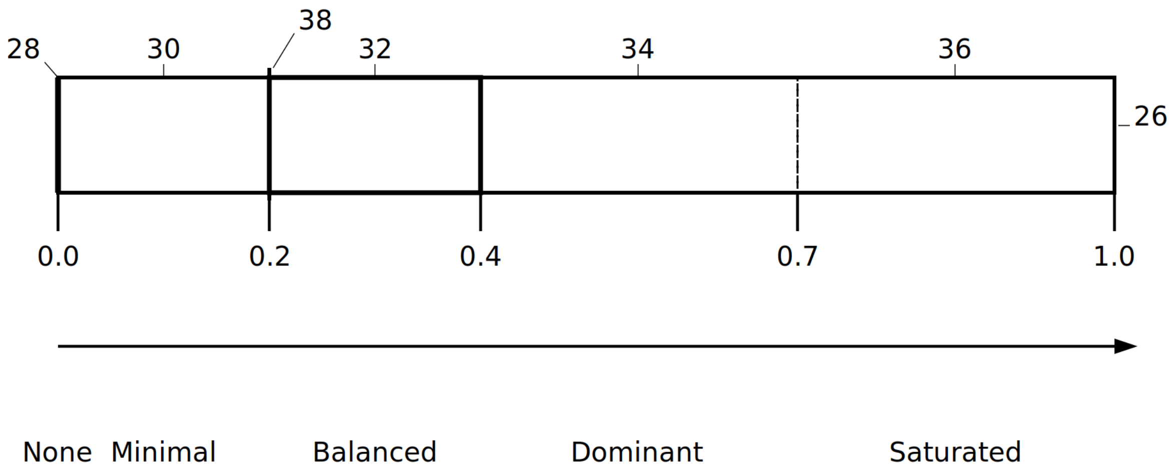

28
30
38
32
34
36
26
0.0
0.2
0.4
0.7
1.0
None
Minimal
Balanced
Dominant
Saturated

FIG. 2

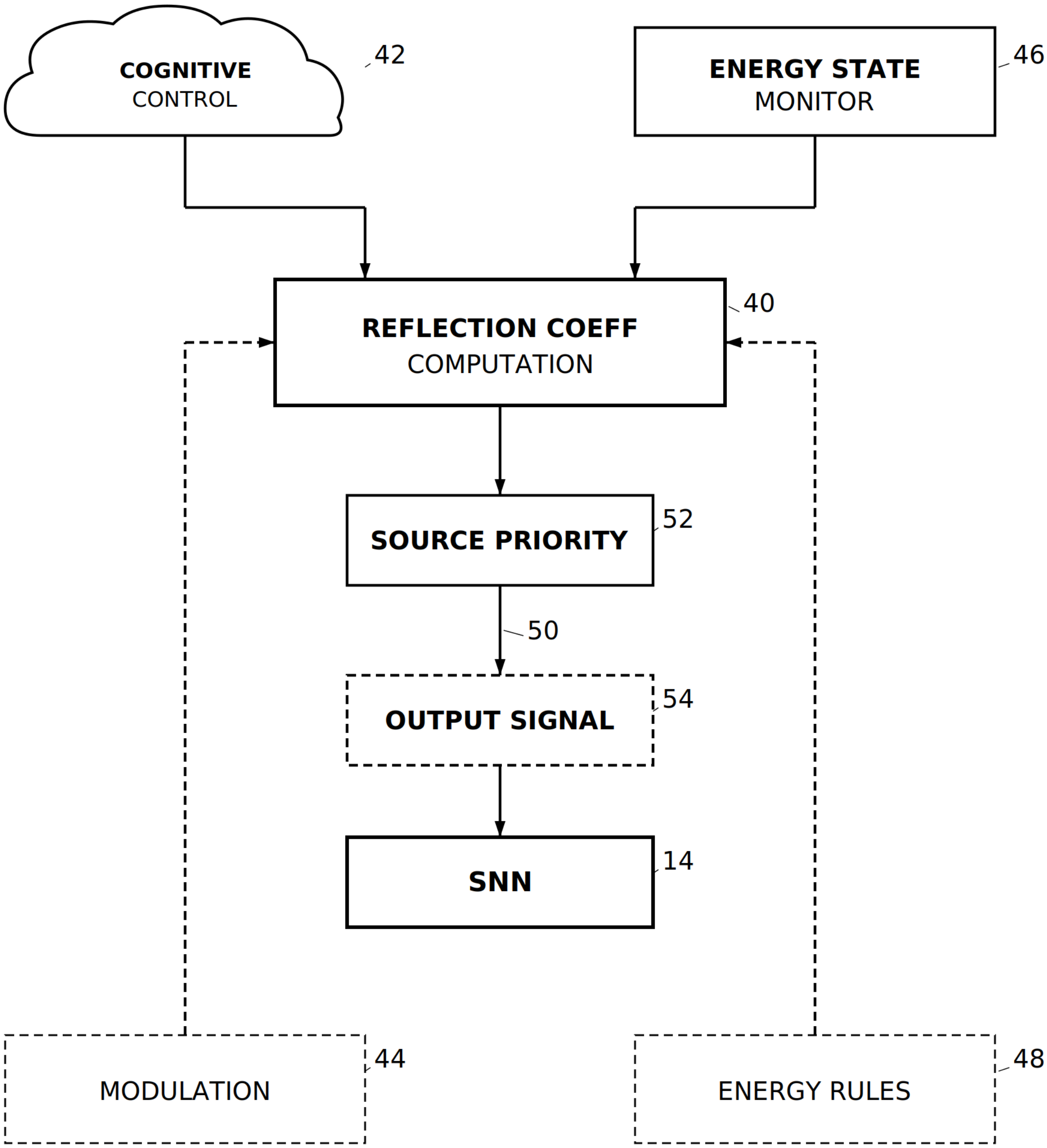

COGNITIVE CONTROL
42
ENERGY STATE MONITOR
46
REFLECTION COEFF COMPUTATION
40
SOURCE PRIORITY
52
50
OUTPUT SIGNAL
54
SNN
14
MODULATION
44
ENERGY RULES
48

FIG. 3

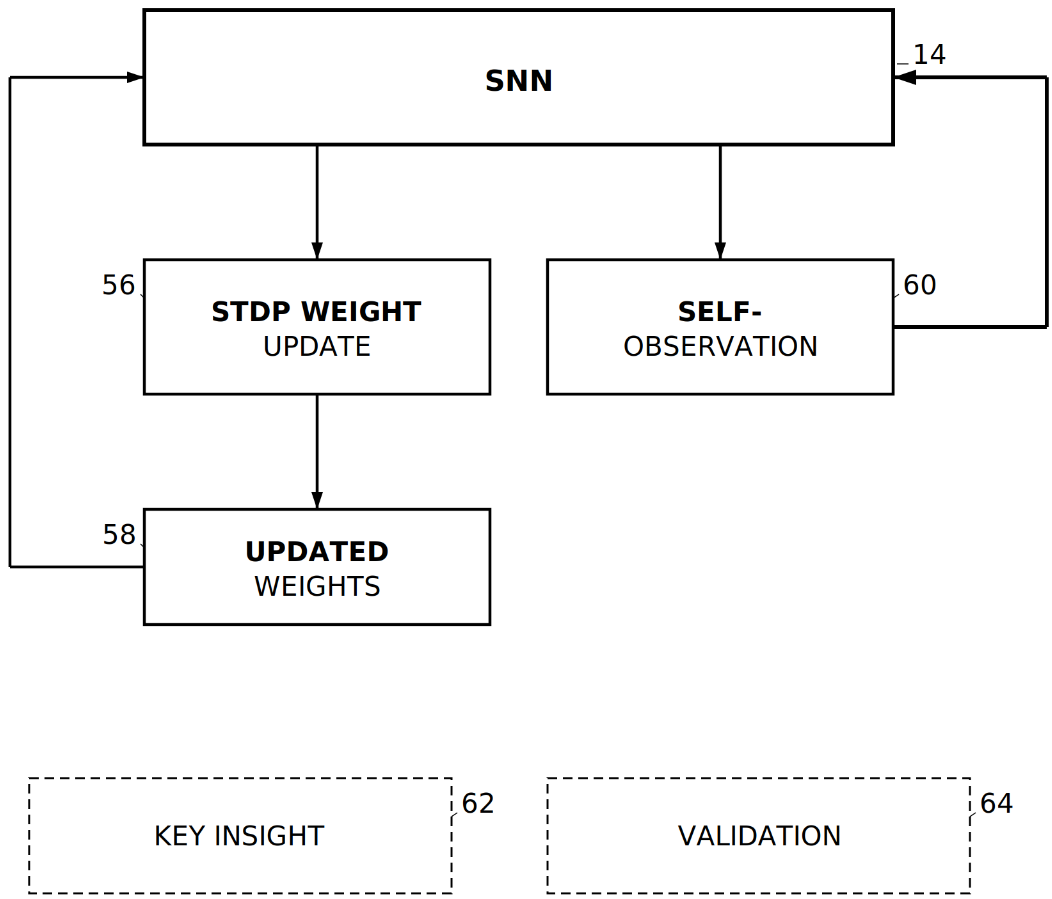

SNN
14
56
STDP WEIGHT
UPDATE
SELF-
OBSERVATION
60
58
UPDATED
WEIGHTS
KEY INSIGHT
62
VALIDATION
64

FIG. 4

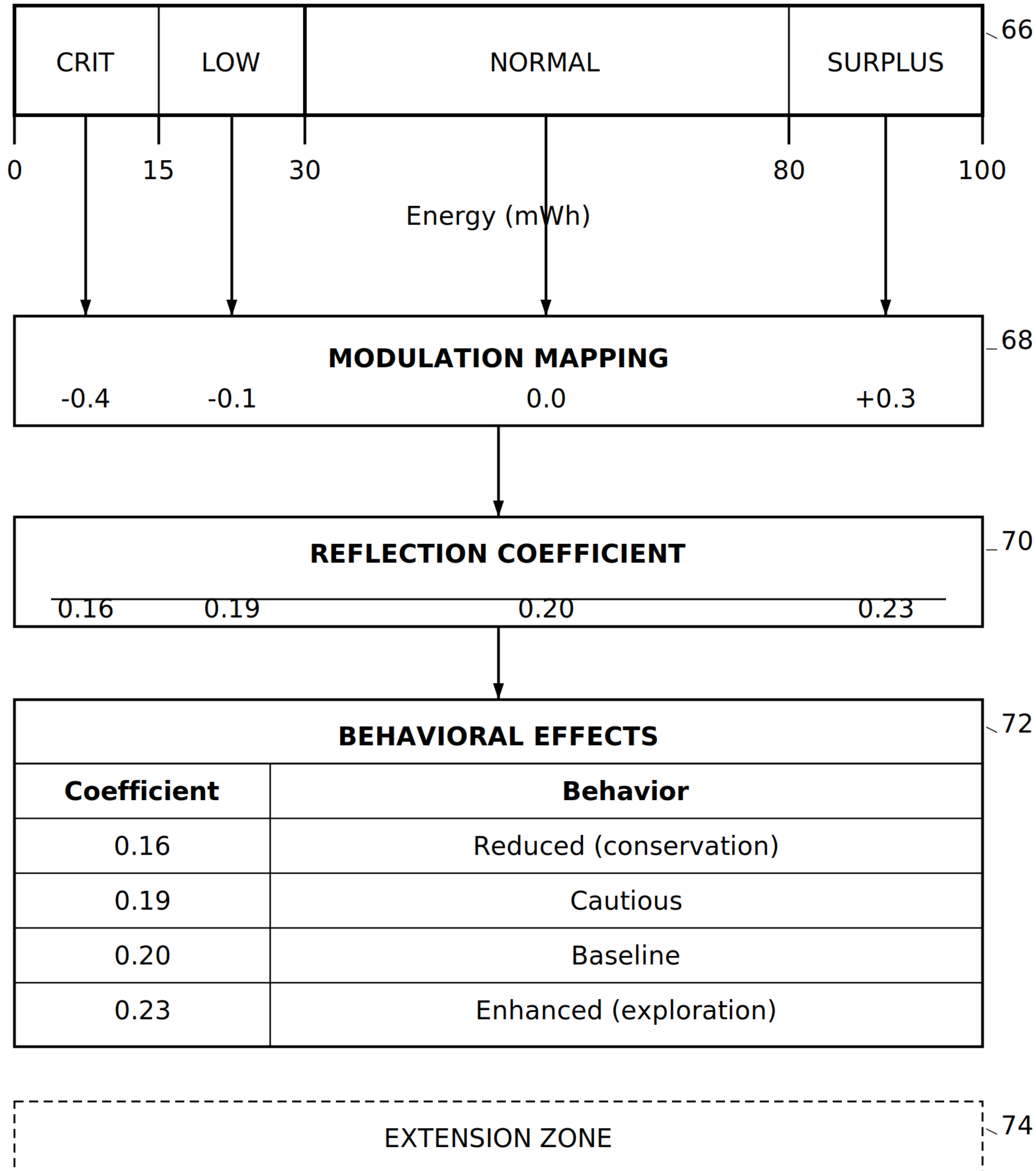
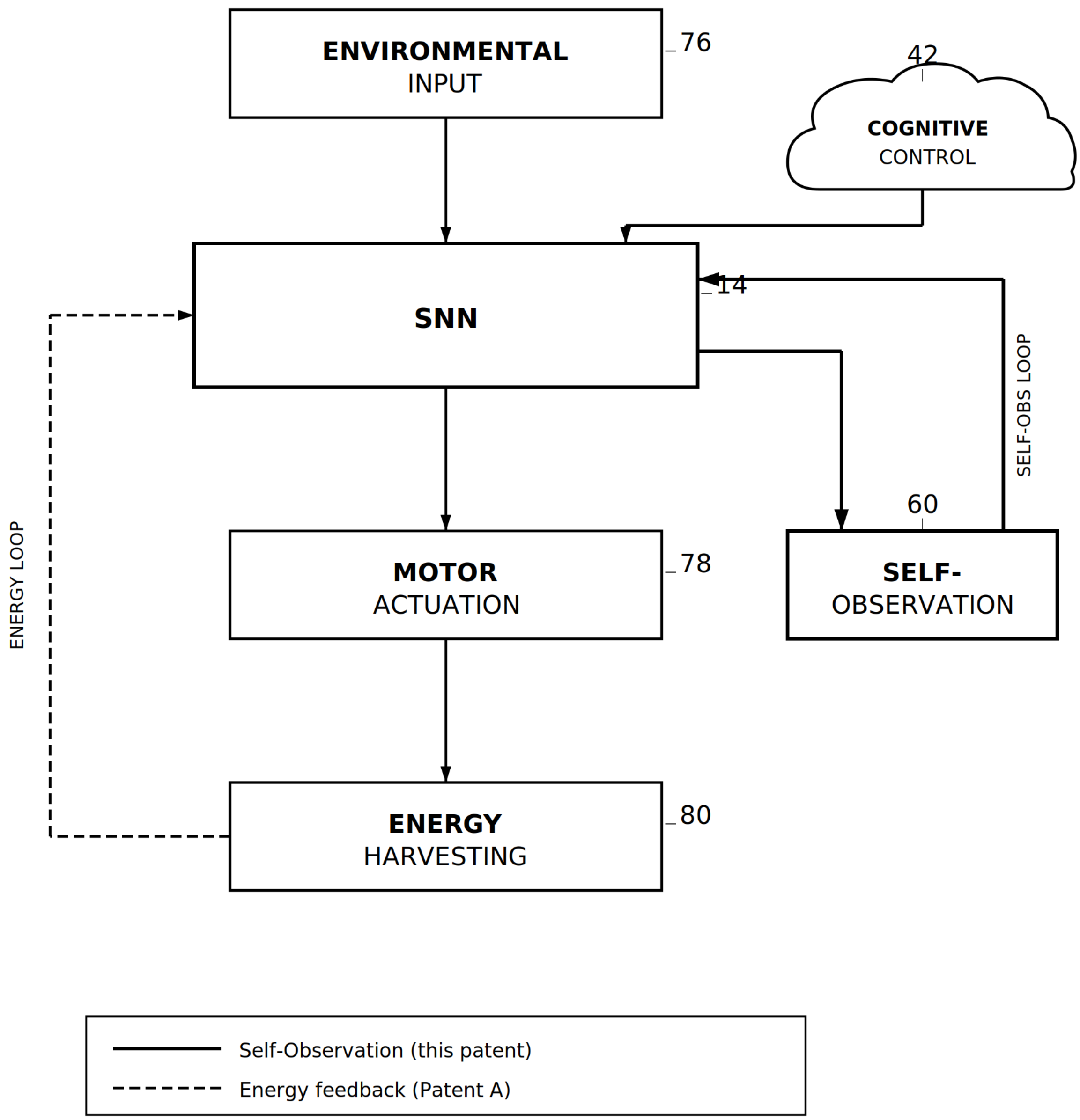

CRIT
LOW
NORMAL
SURPLUS
66
0
15
30
80
100
Energy (mWh)
MODULATION MAPPING
-0.4
-0.1
0.0
+0.3
68
REFLECTION COEFFICIENT
0.16
0.19
0.20
0.23
70
BEHAVIORAL EFFECTS
72
Coefficient
Behavior
0.16
Reduced (conservation)
0.19
Cautious
0.20
Baseline
0.23
Enhanced (exploration)
EXTENSION ZONE
74

FIG. 5



ENVIRONMENTAL
INPUT
76
42
COGNITIVE
CONTROL
SNN
14
SELF-OBS LOOP
ENERGY LOOP
60
MOTOR
ACTUATION
78
SELF-
OBSERVATION
ENERGY
HARVESTING
80
Self-Observation (this patent)
Energy feedback (Patent A)

FIG. 6

PATENT C

onomous Self-Regulation■and Cognitive Resynchronization in Neural Processing■Systems During Disconnection fr

7 drawing sheets

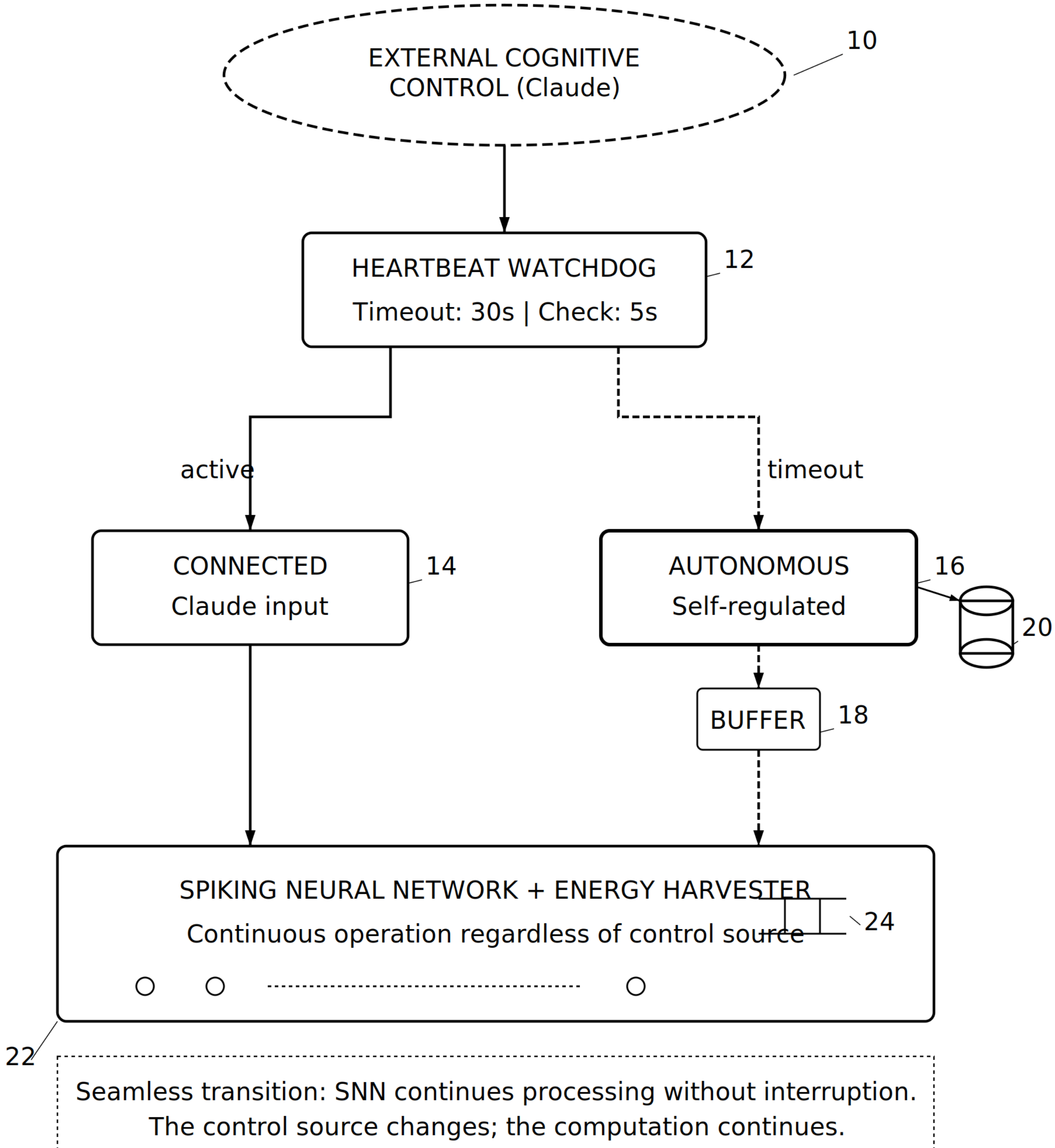
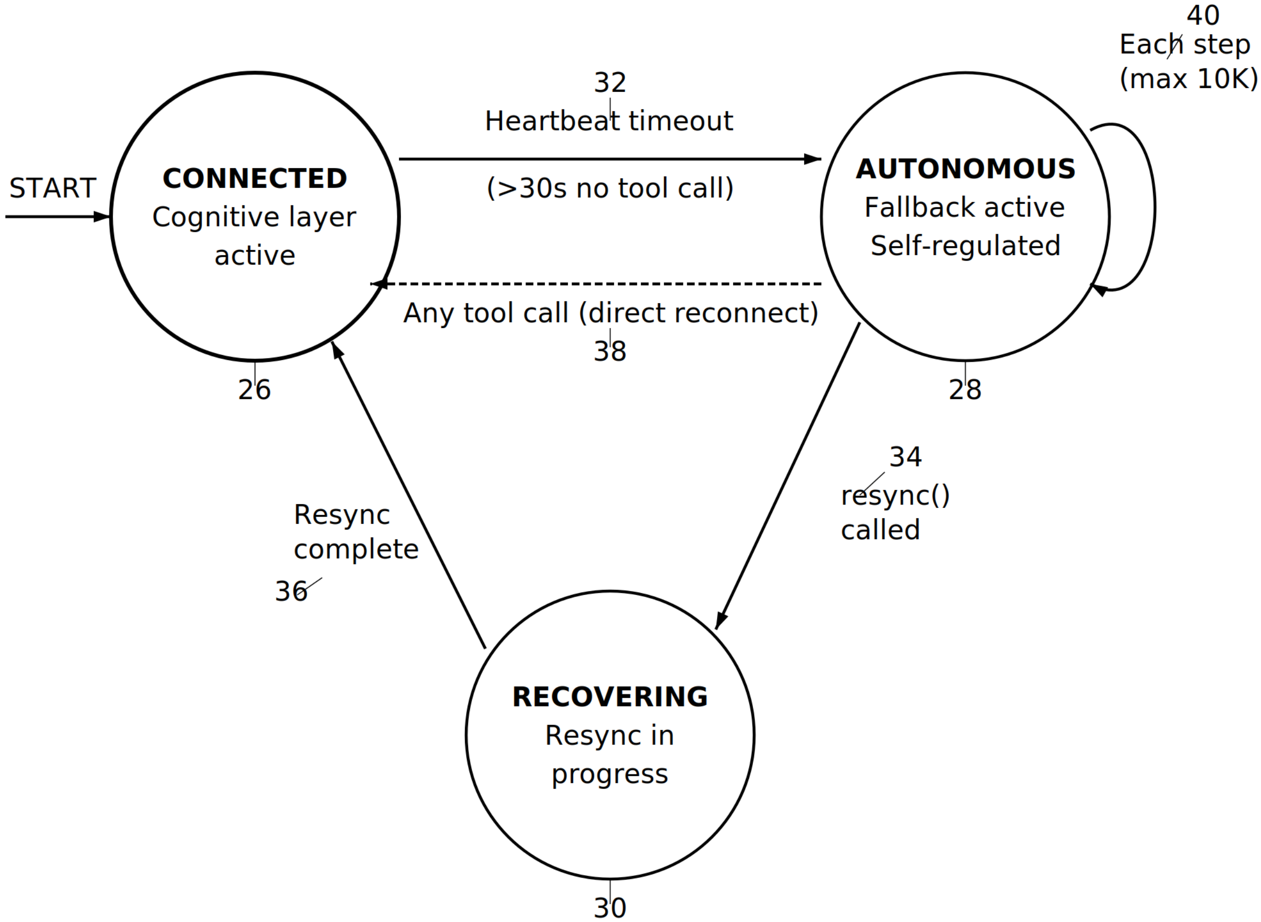

EXTERNAL COGNITIVE
CONTROL (Claude)
10
HEARTBEAT WATCHDOG
Timeout: 30s | Check: 5s
12
active
timeout
CONNECTED
Claude input
14
AUTONOMOUS
Self-regulated
16
20
BUFFER
18
SPIKING NEURAL NETWORK + ENERGY HARVESTER
Continuous operation regardless of control source
24
22
Seamless transition: SNN continues processing without interruption.
The control source changes; the computation continues.

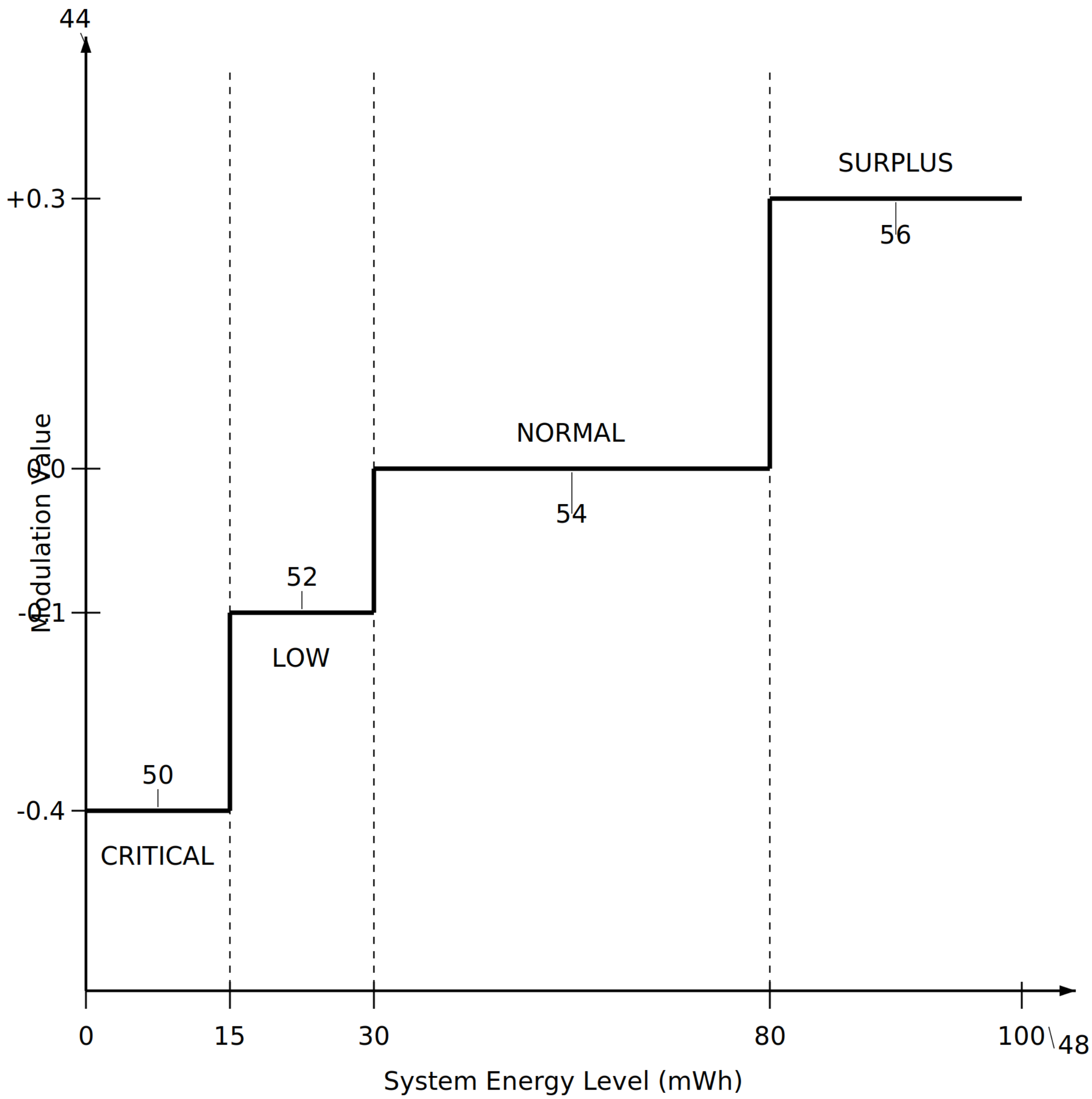
FIG. 1



State Transition Legend: 42

26 CONNECTED: Cognitive layer provides input/modulation
28 AUTONOMOUS: Self-generated input, energy-aware modulation
30 RECOVERING: Buffer transmitted, restoring control

FIG. 2



Behavioral Effects:

CRITICAL (-0.4): Maximum conservation - inhibit neural activity
LOW (-0.1): Cautious operation - slightly reduce activity
NORMAL (0.0): Standard autonomous processing
SURPLUS (+0.3): Opportunistic exploration - utilize excess

46

FIG. 3

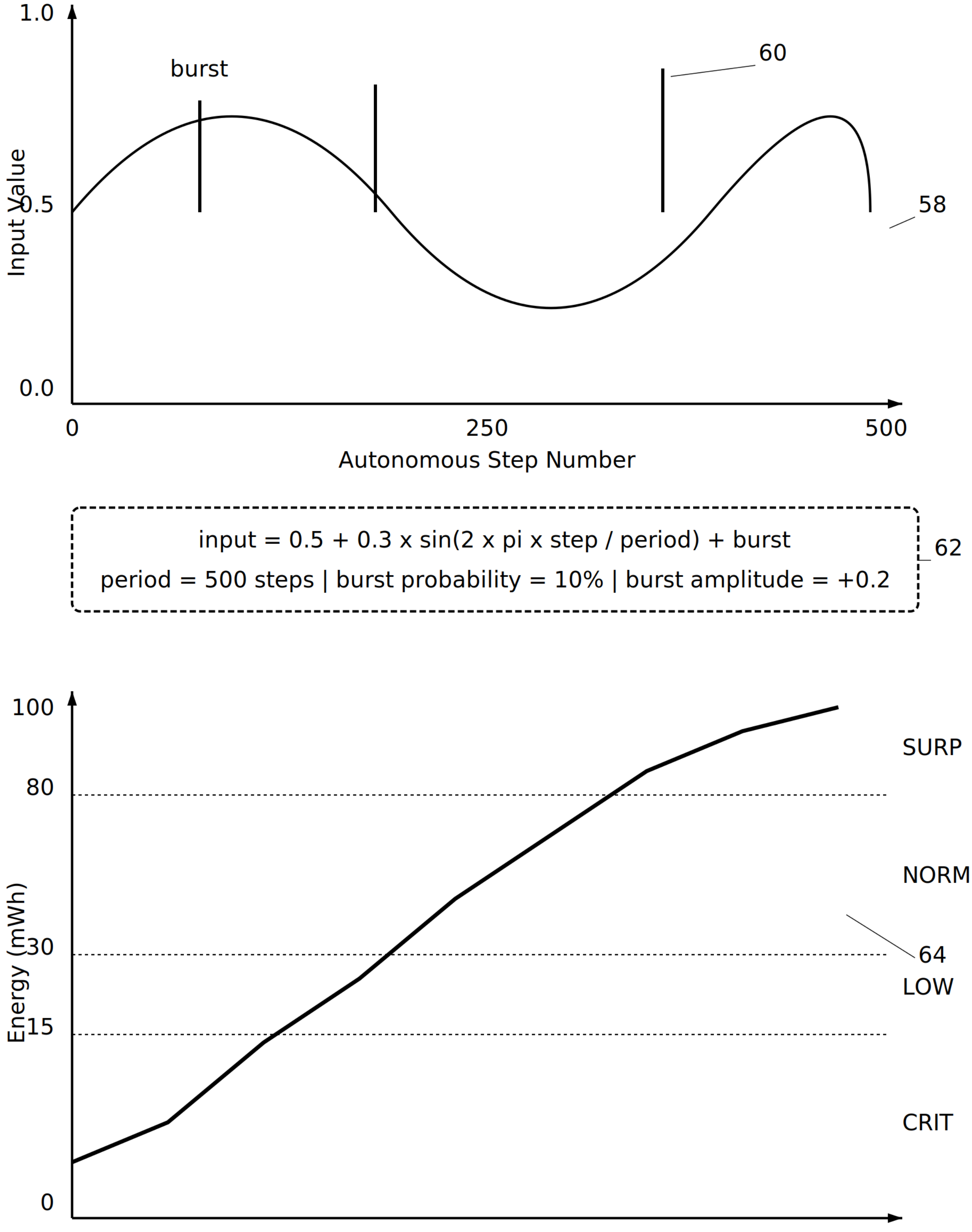

1.0
burst
60
Input Value
0.5
58
0.0
0
250
500
Autonomous Step Number
input = 0.5 + 0.3 x sin(2 x pi x step / period) + burst
period = 500 steps | burst probability = 10% | burst amplitude = +0.2
62
100
SURP
80
NORM
Energy (mWh)
30
64
LOW
15
CRIT
0
Autonomous Step Number

FIG. 4

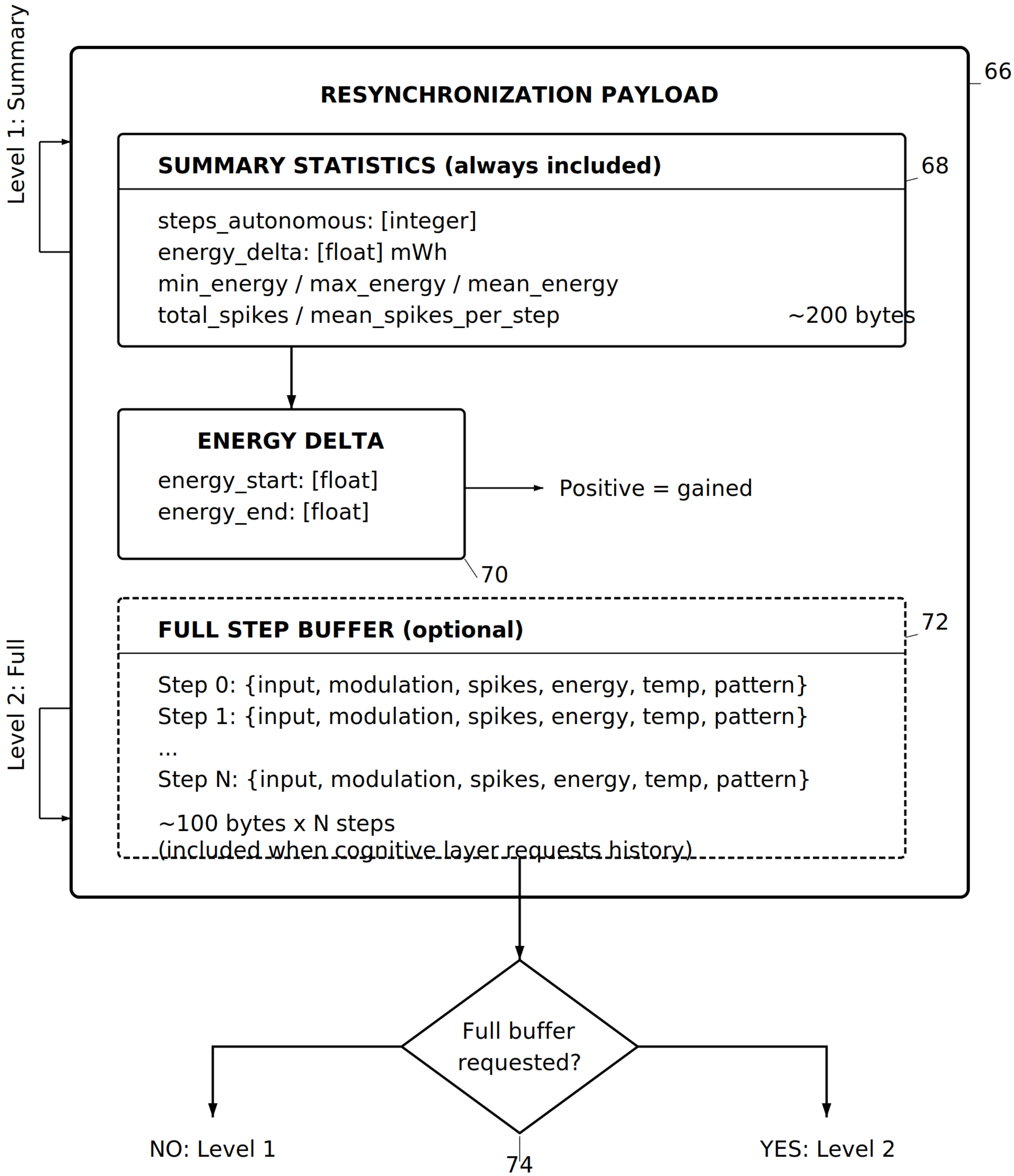

Level 1: Summary
RESYNCHRONIZATION PAYLOAD
66
SUMMARY STATISTICS (always included)
68
steps_autonomous: [integer]
energy_delta: [float] mWh
min_energy / max_energy / mean_energy
total_spikes / mean_spikes_per_step
~200 bytes
ENERGY DELTA
energy_start: [float]
energy_end: [float]
Positive = gained
70
Level 2: Full
FULL STEP BUFFER (optional)
72
Step 0: {input, modulation, spikes, energy, temp, pattern}
Step 1: {input, modulation, spikes, energy, temp, pattern}
...
Step N: {input, modulation, spikes, energy, temp, pattern}
~100 bytes x N steps
(included when cognitive layer requests history)
Full buffer requested?
NO: Level 1
74
YES: Level 2

FIG. 5

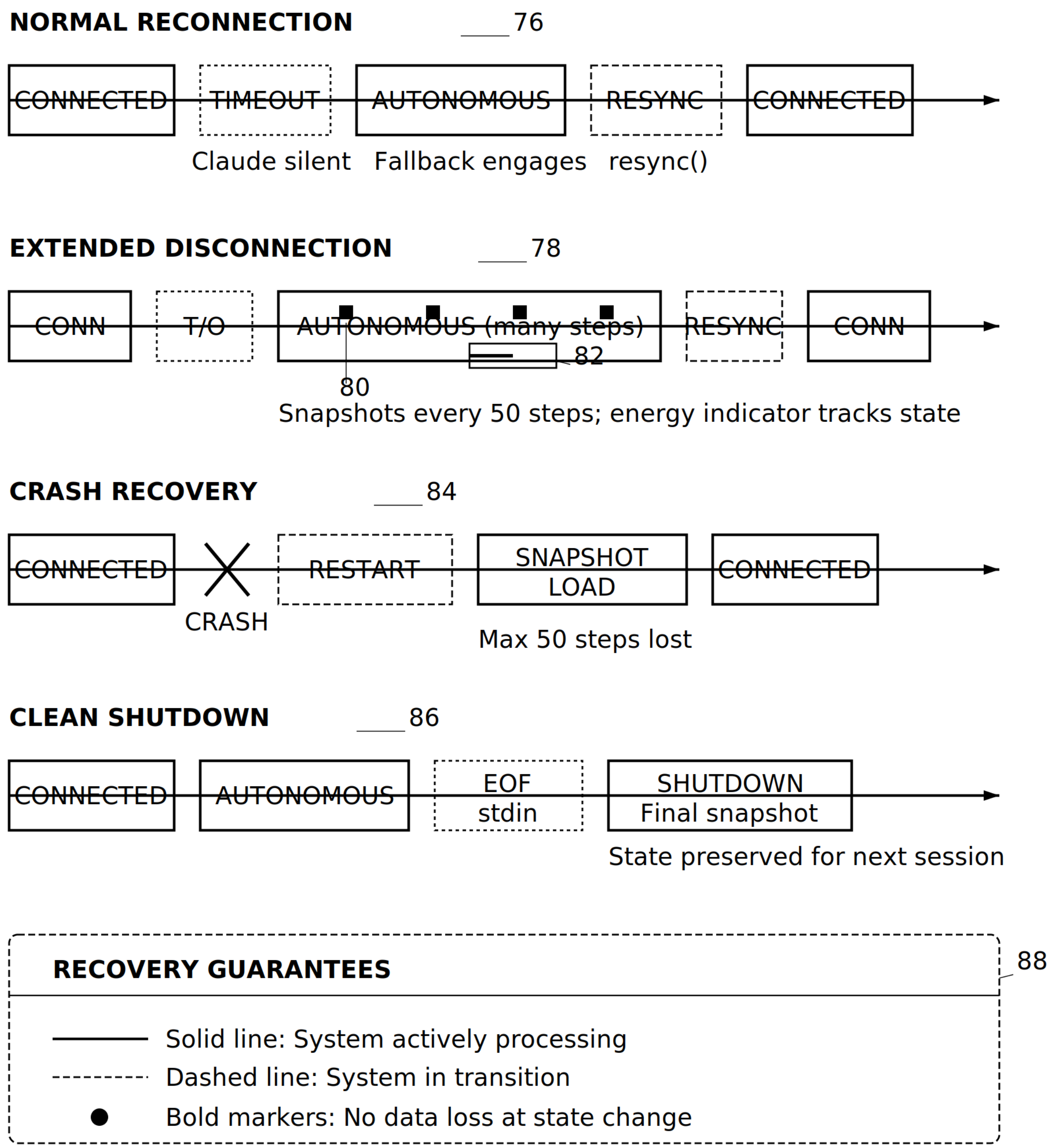
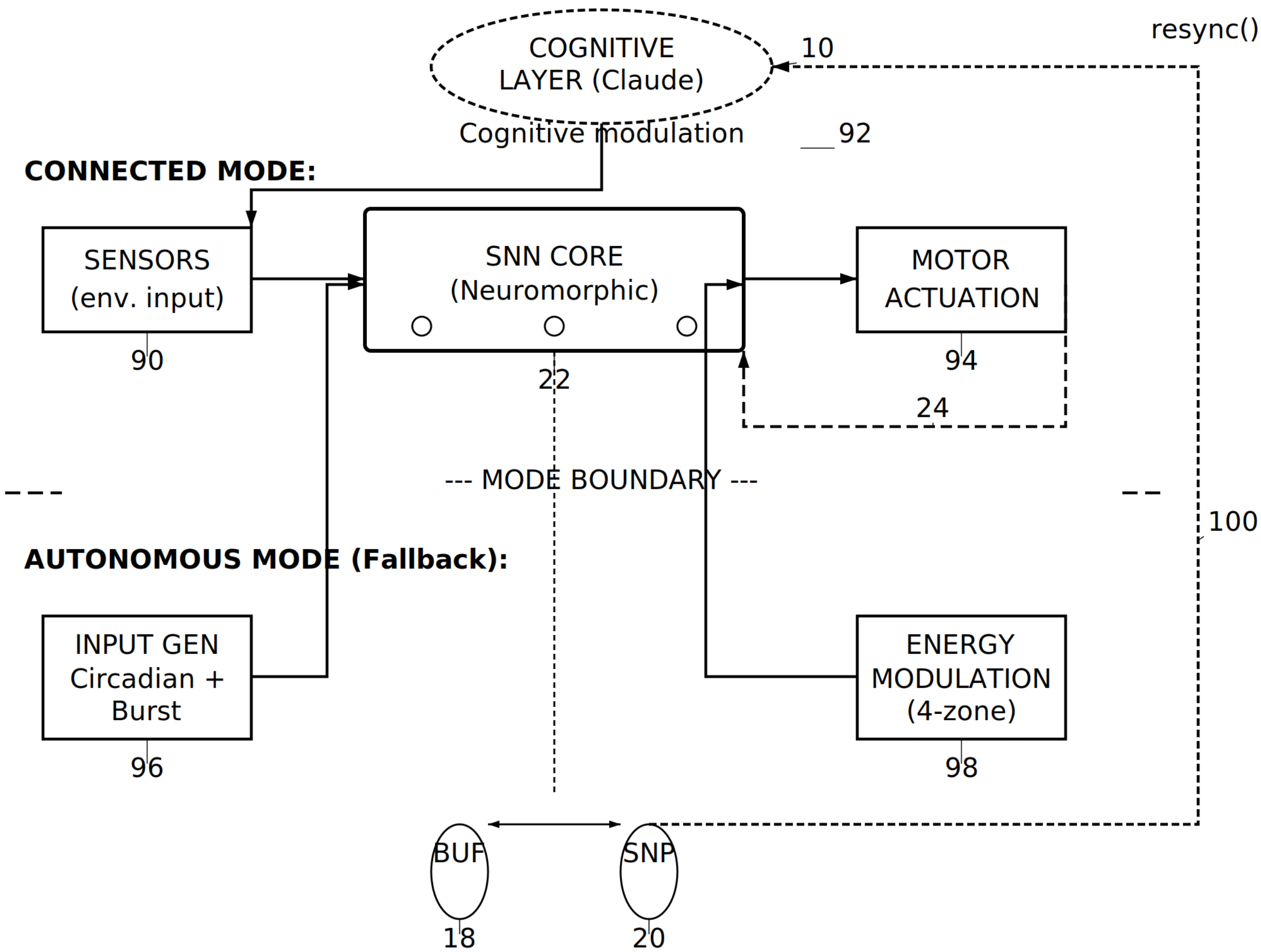

NORMAL RECONNECTION
76
CONNECTED
TIMEOUT
AUTONOMOUS
RESYNC
CONNECTED
Claude silent
Fallback engages
resync()
EXTENDED DISCONNECTION
78
CONN
T/O
AUTONOMOUS (many steps)
RESYNC
CONN
82
80
Snapshots every 50 steps; energy indicator tracks state
CRASH RECOVERY
84
CONNECTED
RESTART
SNAPSHOT
LOAD
CONNECTED
CRASH
Max 50 steps lost
CLEAN SHUTDOWN
86
CONNECTED
AUTONOMOUS
EOF
stdin
SHUTDOWN
Final snapshot
State preserved for next session
RECOVERY GUARANTEES
88
Solid line: System actively processing
Dashed line: System in transition
Bold markers: No data loss at state change

FIG. 6

END-TO-END SIGNAL FLOW: CONNECTED vs AUTONOMOUS



MODE SELECTION LOGIC:

CONNECTED: Input from sensors (90), modulation from Claude (92)
AUTONOMOUS: Input from generator (96), modulation from energy (98)
Buffer (18) and snapshot (20) active during autonomous mode
Feedback path (100) transmits resync payload on reconnection

FIG. 7